

Deflection of beams

Introduction

Deflection of Beams (Solution Method by Direct Integration)

Moment - Area Method for finding Beam Deflections



Introduction

The axis of a beam deflects from its initial position under action of applied forces. Accurate values for these beam deflections are sought in many practical cases: elements of machines must be sufficiently rigid to prevent misalignment and to maintain dimensional accuracy under load; in buildings, floor beams cannot deflect excessively to avoid the undesirable psychological effect of flexible floors on occupants and to minimize or prevent distress in brittle-finish materials; likewise, information on deformation characteristics of members is essential in the study of vibrations of machines as well as of stationary and flight structures.

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Deflection of Beams

(Solution Method by Direct Integration)

From Analytic geometry, Curvature of a line,

$$k = \frac{1}{R} = \frac{\frac{d^2y}{dx^2}}{\left\{1 + \left(\frac{d^2y}{dx^2}\right)^2\right\}^{3/2}}$$

where x and y are co-ordinates of a point on the curve.

For small deflections,

$$k = \frac{d^2y}{dx^2}$$

Since, $\sigma_x = -My / EI$

and $k = -\varepsilon / y$, and $\varepsilon = \sigma_x / E$

$$k = M / EI$$

$$\therefore M = EI \frac{d^2y}{dx^2}$$

where $M = M_y$

Hence,

$$EI \frac{d^2y}{dx^2} = M(x)$$

$$\text{Since, } V = \frac{dM}{dx}$$

$$EI \frac{d^3y}{dx^3} = V(x)$$

$$\text{Since, } w = \frac{dv}{dx}$$

$$EI \frac{d^4 y}{dx^4} = w(x)$$

Boundary Conditions

Refer figure 5.2.7(a) – (d)

(a) Clamped Support:

$$y(x_1) = 0; y'(x_1) = 0;$$

(b) Roller or Pinned Support:

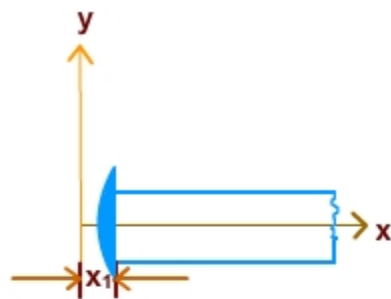
$$y(x_1) = 0; M(x_1) = 0;$$

(c) Free end:

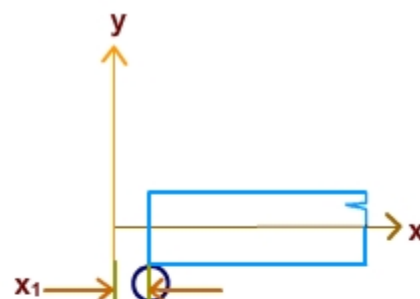
$$M(x_1) = 0; V(x_1) = 0;$$

(d) Guided Support:

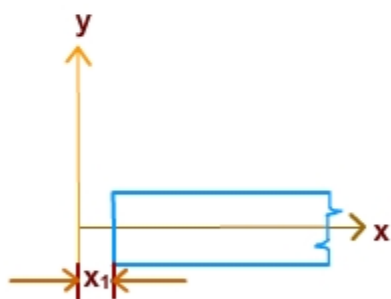
$$y'(x_1) = 0; V(x_1) = 0;$$



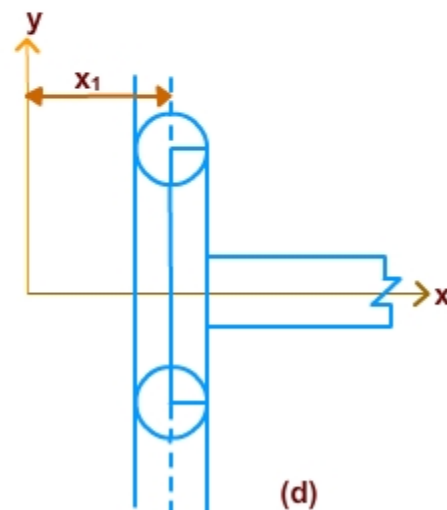
(a)



(b)



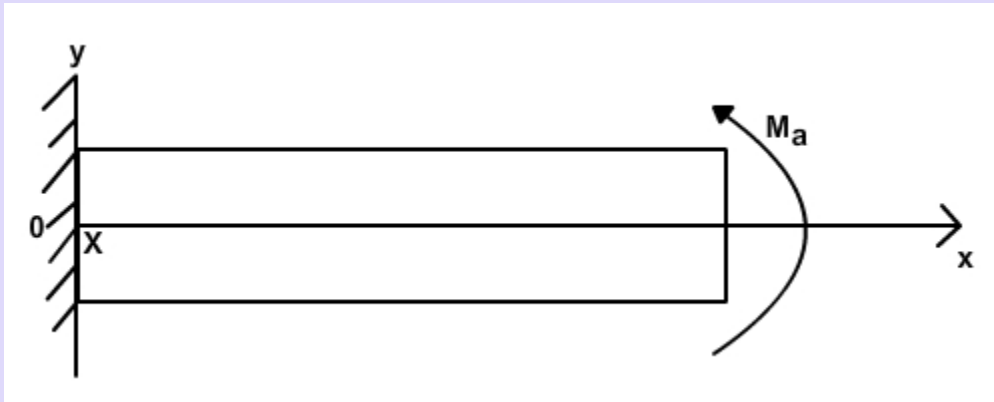
(c)



(d)

Example:

Question: A Cantilever beam is subjected to a bending moment M at the free end. Take flexural rigidity to be constant and equal to EI . Find the equation of the elastic curve.



$$EI \frac{d^2 y}{dx^2} = M_a$$

Integrating

$$EI \frac{dy}{dx} = M_a x + C_1$$

$$\text{at } x = 0; \frac{dy}{dx} = 0$$

$$\text{which gives } C_1 = 0$$

Integrating again,

$$EI y = \frac{M_a x^2}{2} + C_2$$

$$y = 0; \text{ at } x = 0 \text{ gives}$$

$$C_2 = 0$$

$$\therefore y = \frac{M_a x^2}{2EI}$$

which is the equation to the elastic curve.

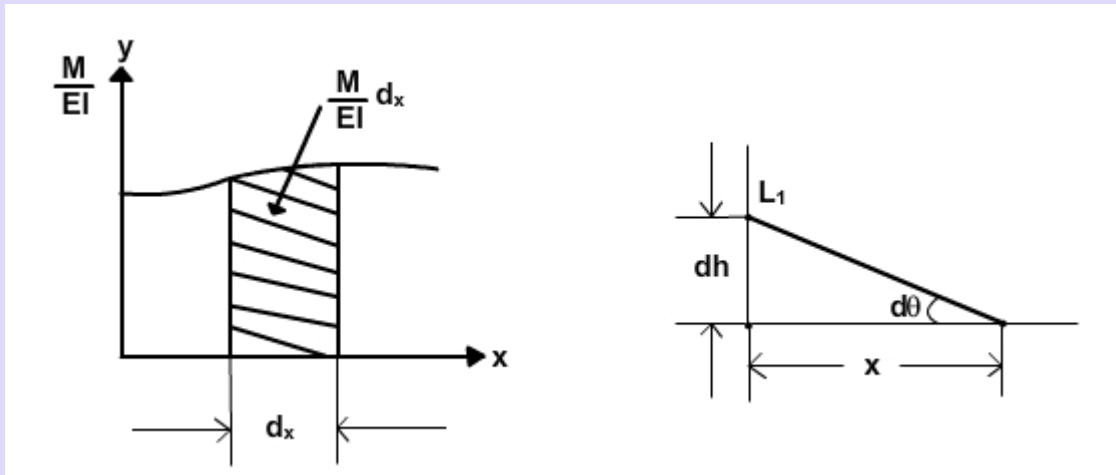
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Moment - Area Method for finding Beam Deflections

This method is used generally to obtain displacement and rotation at a single point on a beam.

This method makes use of the Moment - Area theorems given below.

Moment - Area Theorems



Refer above Figure

$$\frac{M}{EI} = \frac{d^2 y}{dx^2}$$

$$\frac{M}{EI} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$$

Let $dy/dx = \theta$

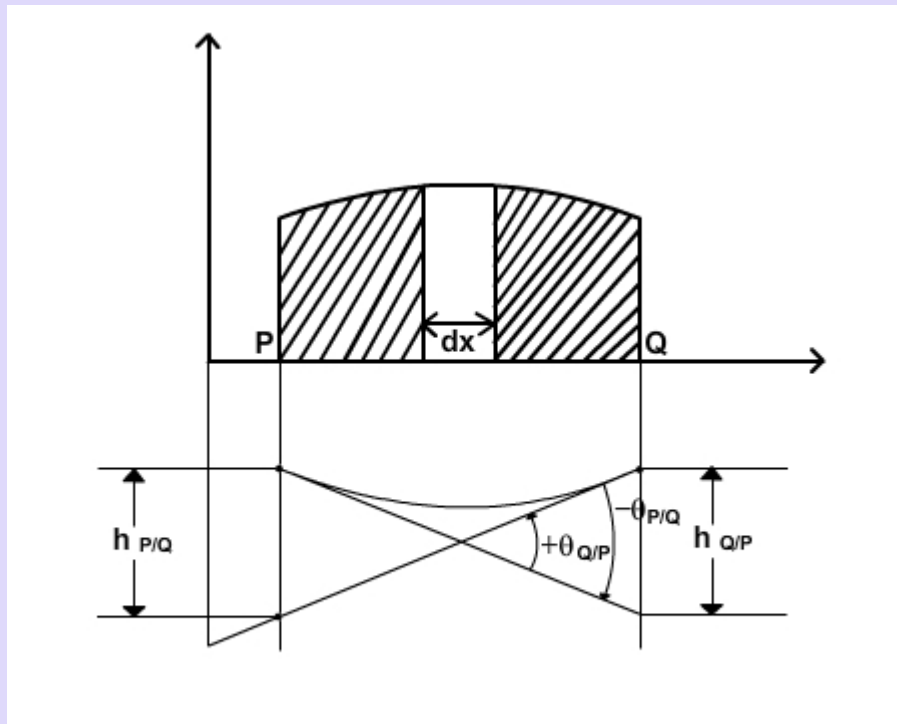
$$\frac{M}{EI} = \frac{d\theta}{dx}$$

$$\int d\theta = \int \frac{M}{EI} dx$$

Referring to figure down

$$\therefore \theta_{Q/P} = \theta_Q - \theta_P = \int_P^Q \frac{M}{EI} dx$$

This is the first moment area theorem, Where P and Q are any two sections on the beam. ie change in angle measured in radians between any two point P and Q on the elastic curve is equal to the M/EI area bounded by the ordinates through P and Q.



Referring to Figure, considering an element of the Elastic Curve,

$$dh = x \cdot d\theta$$

$$\begin{aligned} \therefore h_{Q/P} &= \int_P^Q dh \\ &= \int_P^Q x d\theta \\ &= \int_P^Q \frac{M}{EI} x dx \end{aligned}$$

This is the second moment Area theorem.

If $\bar{x}_P$ is the distance of centroid of the bending moment diagram between P and Q from P,

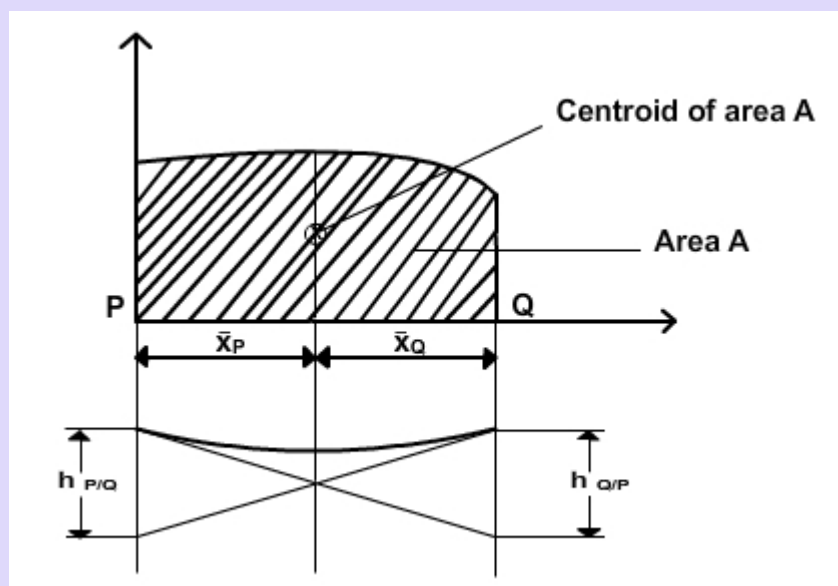
(Refer Figure)

then

$$h_{P/Q} = A \bar{x}_P$$

$$h_{Q/P} = A \bar{x}_Q$$

Here $h_{P/Q}$ is called the tangent deviation of the point P from a tangent at Q.



where A is the area of the BM diagram between P and Q . $\bar{x}_P$ and $\bar{x}_Q$ are as shown in figure.

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